

ACTIVIDAD

4

ESTRUCTURA DE BASE DE DATOS



Integrantes:

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Objetivo:

El objetivo desarrollar un programa en Java que simule un sistema de gestión de tareas para una pequeña empresa. Este sistema utilizará diferentes estructuras de datos (pilas, colas y listas) para manejar tareas asignadas a distintos departamentos. (Practicar la implementación de estas estructuras y entender sus aplicaciones prácticas).

Diagramas de flujo Cliente:

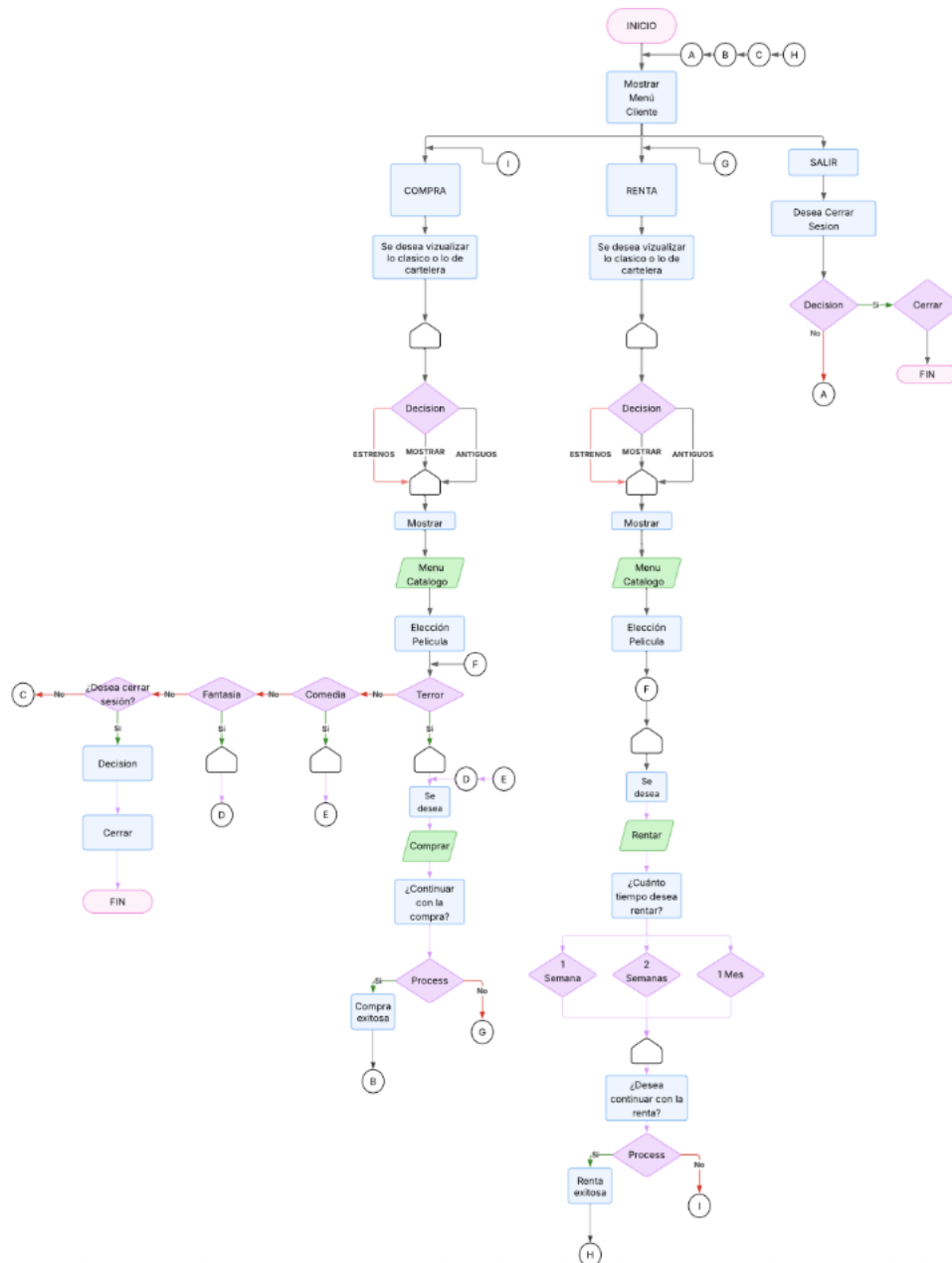


Diagrama de flujo Empleado:

Los clientes pueden agregar películas a su carrito, ver su carrito, ver las películas disponibles, eliminar películas del carrito o agregarlas; mientras que los trabajadores son

capaces de ver sus tareas pendientes, agregar o eliminar sus tareas y dividir las tareas dependiendo de su nivel de prioridad.

Código:

```
package com.example.UI;

import com.example.process.Manager;

import java.awt.*;
import java.awt.event.ActionListener;
import java.awt.event.*;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JOptionPane;
import javax.swing.JPanel;
import javax.swing.JScrollPane;
import javax.swing.JTextArea;
import javax.swing.JTextField;

import com.example.data.Peliculas;
```

```
public class CLI {
    private Manager manager = new Manager();

    public void loginAdmin(){
        JFrame frame = new JFrame(title:"Inicio");
        frame.setSize(width:600, height:400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(manager:null);
        frame.getContentPane().setBackground(Color.LIGHT_GRAY);

        Font fontGrande = new Font(name:"Arial", Font.BOLD, size:30);
        Font fontTitulo = new Font(name:"Arial", Font.PLAIN, size:25);
        Font fontNormal = new Font(name:"Arial", Font.PLAIN, size:15);

        JLabel label = new JLabel(text:"CINEMANIA");
        label.setBounds(x:110, y:20, width:300, height:30);
        label.setFont(fontGrande);
        label.setForeground(Color.white);
        frame.add(label);

        JLabel label2 = new JLabel(text:"Administrador");
        label2.setBounds(x:240, y:90, width:200, height:30);
        label2.setFont(fontTitulo);
        label2.setForeground(Color.black);
        frame.add(label2);
    }
}
```

```
JPanel panel2 = new JPanel(layout:null);
panel2.setBounds(x:0, y:0, width:90, height:600);
panel2.setBackground(Color.red);
frame.add(panel2);

JPanel panel = new JPanel(layout:null);
panel.setBounds(x:0,y:0,width:600,height:70);
panel.setBackground(Color.blue);
frame.add(panel);

JButton botonAdmin = new JButton(text:"Admin");
botonAdmin.setBounds(x:5, y:90, width:80, height:25);
panel2.add(botonAdmin);

JButton botonUser = new JButton(text:"User");
botonUser.setBounds(x:5, y:180, width:80, height:25);
panel2.add(botonUser);

JButton botonWorker = new JButton(text:"Worker");
botonWorker.setBounds(x:5, y:270, width:80, height:25);
panel2.add(botonWorker);

JLabel labelUser = new JLabel(text:"Usuario:");
labelUser.setBounds(x:165, y:150, width:100, height:30);
labelUser.setForeground(Color.black);
frame.add(labelUser);
```

```
JLabel labelPass = new JLabel(text:"Contraseña:");
labelPass.setBounds(x:150, y:200, width:100, height:30);
labelPass.setForeground(Color.black);
frame.add(labelPass);

JTextField user = new JTextField();
user.setBounds(x:250, y:150, width:200, height:30);
frame.add(user);

JTextField pass = new JTextField();
pass.setBounds(x:250, y:200, width:200, height:30);
frame.add(pass);

JButton botonAceptar = new JButton(text:"Aceptar");
botonAceptar.setBounds(x:300, y:250, width:100, height:30);
frame.add(botonAceptar);
```

```
botonAceptar.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = user.getText();
        String password = pass.getText();
        if(username.isEmpty() || password.isEmpty()) {
            JOptionPane.showMessageDialog(frame, message:"Los datos no pueden estar vacios.");
        } else {
            if (!username.equals(anObject:"admin") || !password.equals(anObject:"admin123")) {
                JOptionPane.showMessageDialog(frame, message:"Usuario o contraseña incorrecta.");
            } else {
                frame.dispose();
                windowAdmin();
            }
        }
    }
});
```

```
        botonUser.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e){
                frame.dispose();
                CLI_Client cli_Client = new CLI_Client();
                cli_Client.loginUser();
            }
        });

        botonWorker.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                CLI_Worker cli_Worker = new CLI_Worker();
                cli_Worker.loginWorker();
            }
        });

        frame.setVisible(b:true);
    }
```

```
public void windowAdmin(){
    JFrame frameAdmin = new JFrame(title:"Administracion");
    frameAdmin.setSize(width:600, height:400);
    frameAdmin.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frameAdmin.setLocationRelativeTo(c:null);

    JPanel panel = new JPanel(new BorderLayout());

    JTextArea textArea = new JTextArea();
    textArea.setEditable(b:false);

    JScrollPane scrollPane = new JScrollPane(textArea);
    panel.add(scrollPane, BorderLayout.CENTER);

    JPanel buttJPanel = new JPanel(new GridLayout(rows:4, cols:1, hgap:10, vgap:10));

    JButton btnAgregar = new JButton(text:"Agregar película");
    JButton btnMostrar = new JButton(text:"Mostrar películas");
    JButton btnEliminar = new JButton(text:"Eliminar película");
    JButton btnSalir = new JButton(text:"Salir");

    buttJPanel.add(btnAgregar);
    buttJPanel.add(btnMostrar);
    buttJPanel.add(btnEliminar);
    buttJPanel.add(btnSalir);
```

```
    panel.add(buttJPanel, BorderLayout.EAST);

    frameAdmin.add(panel);

    btnAgregar.addActionListener(e -> agregarPelicula());
    btnMostrar.addActionListener(e -> menuMostrar());
    //btnEliminar.addActionListener(e -> eliminarPelicula());
    btnSalir.addActionListener(e -> System.exit(status:0));

    frameAdmin.setVisible(b:true);
}
```

```
private void agregarPelicula() {
    String titulo = JOptionPane.showInputDialog(this, initialSelectionValue:"Ingrese el titulo:");
    String genero = JOptionPane.showInputDialog(this, initialSelectionValue:"Ingrese el genero:");
    String añoStr = JOptionPane.showInputDialog(this, initialSelectionValue:"Ingrese el año:");

    try {
        int año = Integer.parseInt(añoStr);
        manager.addPeliculas(titulo, genero, año);
        JOptionPane.showMessageDialog(parentComponent:null, message:"Pelicula agregada con exito.");
    } catch (NumberFormatException ex) {
        JOptionPane.showMessageDialog(parentComponent:null, message:"El año debe ser un numero valido.");
    }
}
```

```
package com.example.UI;

import com.example.data.Peliculas;
import com.example.process.ClientManager;
import javax.swing.*;
import java.awt.Color;
import java.awt.Font;
import java.awt.event.*;
import java.util.*;
```

```
public class CLI_Client {
    private ClientManager clientManager = new ClientManager();

    public void loginUser() {
        JFrame frame = new JFrame(title:"Inicio");
        frame.setSize(width:600, height:400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(manager:null);
        frame.getContentPane().setBackground(Color.LIGHT_GRAY);

        Font fontGrande = new Font(name:"Arial", Font.BOLD, size:30);
        Font fontTitulo = new Font(name:"Arial", Font.PLAIN, size:25);
        Font fontNormal = new Font(name:"Arial", Font.PLAIN, size:15);

        JLabel label = new JLabel(text:"CINEMANIA");
        label.setBounds(x:110, y:20, width:300, height:30);
        label.setFont(fontGrande);
        label.setForeground(Color.white);
        frame.add(label);

        JLabel label2 = new JLabel(text:"Usuario");
        label2.setBounds(x:280, y:90, width:300, height:30);
        label2.setFont(fontTitulo);
        label2.setForeground(Color.BLACK);
        frame.add(label2);
    }
}
```

```
JPanel panel2 = new JPanel(layout:null);
panel2.setBounds(x:0, y:0, width:90, height:600);
panel2.setBackground(Color.red);
frame.add(panel2);

JPanel panel = new JPanel(layout:null);
panel.setBounds(x:0,y:0,width:600,height:70);
panel.setBackground(Color.blue);
frame.add(panel);

JLabel labelUser = new JLabel(text:"Usuario:");
labelUser.setBounds(x:165, y:150, width:100, height:30);
labelUser.setForeground(Color.black);
frame.add(labelUser);

JLabel labelPass = new JLabel(text:"Contraseña:");
labelPass.setBounds(x:150, y:200, width:100, height:30);
labelPass.setForeground(Color.black);
frame.add(labelPass);

JTextField user = new JTextField();
user.setBounds(x:250, y:150, width:200, height:30);
frame.add(user);
```

```
JTextField pass = new JTextField();
pass.setBounds(x:250, y:200, width:200, height:30);
frame.add(pass);

JButton botonAceptar = new JButton(text:"Aceptar");
botonAceptar.setBounds(x:300, y:250, width:100, height:30);
frame.add(botonAceptar);

JButton botonReg = new JButton(text:"Registrarse");
botonReg.setBounds(x:170, y:250, width:110, height:30);
frame.add(botonReg);

JButton botonAdmin = new JButton(text:"Admin");
botonAdmin.setBounds(x:5, y:90, width:80, height:25);
panel2.add(botonAdmin);

JButton botonUser = new JButton(text:"User");
botonUser.setBounds(x:5, y:180, width:80, height:25);
panel2.add(botonUser);

JButton botonWorker = new JButton(text:"Worker");
botonWorker.setBounds(x:5, y:270, width:80, height:25);
panel2.add(botonWorker);
```



```
botonAceptar.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = user.getText();
        String password = pass.getText();
        if (username.isEmpty() || password.isEmpty()) {
            JOptionPane.showMessageDialog(frame, message:"Los campos no pueden estar vacios.");
        } else {
            if (!clientManager.login(username, password)) {
                JOptionPane.showMessageDialog(frame, message:"Usuario o contraseña incorrectos.");
            } else {
                frame.dispose();
                //windowUser();
            }
        }
    }
});
```

```
botonReg.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = user.getText();
        String password = pass.getText();
        if (username.isEmpty() || password.isEmpty()) {
            JOptionPane.showMessageDialog(frame, message:"Los campos no pueden estar vacios.");
        } else {
            clientManager.addClient(username, password);
            JOptionPane.showMessageDialog(frame, message:"Usuario registrado exitosamente.");
        }
    }
});
```

```
botonAdmin.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        frame.dispose();
        CLI cli = new CLI();
        cli.loginAdmin();
    }
});

botonWorker.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        CLI_Worker cli_Worker = new CLI_Worker();
        cli_Worker.loginWorker();
    }
});

frame.setVisible(b:true);
}
```

```
package com.example.UI;

import java.util.*;
import com.example.data.Task;
import com.example.process.WorkerManager;
import javax.swing.*;
import java.awt.Color;
import java.awt.Font;
import java.awt.event.*;
import com.example.process.*;
```

```
public class CLI_Worker {
    public void loginWorker() {
        WorkerManager workerManager = new WorkerManager();

        JFrame frame = new JFrame(title:"Inicio");
        frame.setSize(width:600, height:400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(manager:null);
        frame.getContentPane().setBackground(Color.LIGHT_GRAY);

        Font fontGrande = new Font(name:"Arial", Font.BOLD, size:30);
        Font fontTitulo = new Font(name:"Arial", Font.PLAIN, size:25);
        Font fontNormal = new Font(name:"Arial", Font.PLAIN, size:15);

        JLabel label = new JLabel(text:"CINEMANIA");
        label.setBounds(x:110, y:20, width:300, height:30);
        label.setFont(fontGrande);
        label.setForeground(Color.white);
        frame.add(label);

        JLabel label2 = new JLabel(text:"Trabajador");
        label2.setBounds(x:280, y:90, width:300, height:30);
        label2.setFont(fontTitulo);
        label2.setForeground(Color.BLACK);
        frame.add(label2);
```

```
JPanel panel2 = new JPanel(layout:null);
panel2.setBounds(x:0, y:0, width:90, height:600);
panel2.setBackground(Color.red);
frame.add(panel2);

JPanel panel = new JPanel(layout:null);
panel.setBounds(x:0,y:0,width:600,height:70);
panel.setBackground(Color.blue);
frame.add(panel);

JLabel labelUser = new JLabel(text:"Usuario:");
labelUser.setBounds(x:135, y:125, width:100, height:30);
labelUser.setForeground(Color.black);
frame.add(labelUser);

JLabel labelId = new JLabel(text:"ID:");
labelId.setBounds(x:170, y:160, width:100, height:30);
labelId.setForeground(Color.black);
frame.add(labelId);

JLabel labelDepar = new JLabel(text:"Departamento:");
labelDepar.setBounds(x:100, y:195, width:100, height:30);
labelDepar.setForeground(Color.BLACK);
frame.add(labelDepar);
```

```
JTextField user = new JTextField();
user.setBounds(x:195, y:132, width:100, height:20);
frame.add(user);

JTextField idField = new JTextField();
idField.setBounds(x:195, y:167, width:100, height:20);
frame.add(idField);

JTextField deparField = new JTextField();
deparField.setBounds(x:195, y:201, width:100, height:20);
frame.add(deparField);

JButton botonAceptar = new JButton(text:"Aceptar");
botonAceptar.setBounds(x:300, y:250, width:100, height:30);
frame.add(botonAceptar);

JButton botonReg = new JButton(text:"Registrarse");
botonReg.setBounds(x:170, y:250, width:110, height:30);
frame.add(botonReg);

JButton botonAdmin = new JButton(text:"Admin");
botonAdmin.setBounds(x:5, y:90, width:80, height:25);
panel2.add(botonAdmin);
```

```
JButton botonUser = new JButton(text:"User");
botonUser.setBounds(x:5, y:180, width:80, height:25);
panel2.add(botonUser);

JButton botonWorker = new JButton(text:"Worker");
botonWorker.setBounds(x:5, y:270, width:80, height:25);
panel2.add(botonWorker);
```

```
botonAceptar.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = user.getText();
        String idText = idField.getText();
        String departmentText = deparField.getText();

        if (username.isEmpty() || idText.isEmpty() || departmentText.isEmpty()) {
            JOptionPane.showMessageDialog(frame, message:"Los campos no pueden estar vacios.");
        } else {
            int id = Integer.parseInt(idText);
            int department = Integer.parseInt(departmentText);
            if (id <= 0 || (department <= 0 && department >= 3)) {
                JOptionPane.showMessageDialog(frame, message:"ID o departamento invalido.");
            } else {
                if (!workerManager.login(username, department, id)) {
                    JOptionPane.showMessageDialog(frame, message:"Usuario no encontrado.");
                } else {
                    frame.dispose();
                    //windowWorker();
                }
            }
        }
    }
});
```

```

botonReg.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = user.getText();
        String idText = idField.getText();
        String departmentText = deparField.getText();

        if (username.isEmpty() || idText.isEmpty() || departmentText.isEmpty()) {
            JOptionPane.showMessageDialog(frame, message: "Los campos no pueden estar vacios.");
        } else {
            int id = Integer.parseInt(idText);
            int department = Integer.parseInt(departmentText);

            if (id <= 0 || (department <= 0 && department >= 3)) {
                JOptionPane.showMessageDialog(frame, message: "El ID o departamento son invalidos.");
            } else {
                workerManager.addWorker(username, department, id);
                JOptionPane.showMessageDialog(frame, message: "Usuario registrado exitosamente.");
            }
        }
    }
});

```

```

botonAdmin.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        CLI cli = new CLI();
        cli.loginAdmin();
    }
});

botonUser.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        CLI_Client cli_Client = new CLI_Client();
        cli_Client.loginUser();
    }
});

frame.setVisible(b:true);
}

```

```

package com.example.UI;

import javax.swing.*;
import java.awt.Color;
import java.awt.Font;
import java.awt.event.*;
import com.example.process.*;

```

```

public class CLI2 {
    private ClientManager clientManager = new ClientManager();
    private Manager manager = new Manager();
    private WorkerManager workerManager = new WorkerManager();

    public void window(){
        JFrame frame = new JFrame(title:"Inicio");
        frame.setSize(width:600, height:400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(manager:null);
        frame.getContentPane().setBackground(Color.LIGHT_GRAY);

        Font fontGrande = new Font(name:"Arial", Font.BOLD, size:30);
        Font fontTitulo = new Font(name:"Arial", Font.PLAIN, size:25);
        Font fontNormal = new Font(name:"Arial", Font.PLAIN, size:15);

        JLabel label = new JLabel(text:"CINEMANIA");
        label.setBounds(x:110, y:20, width:300, height:30);
        label.setFont(fontGrande);
        label.setForeground(Color.white);
        frame.add(label);

        JLabel label2 = new JLabel(text:"¡Bienvenido!");
        label2.setBounds(x:240, y:180, width:300, height:30);
        label2.setFont(fontGrande);
        label2.setForeground(Color.WHITE);
        frame.add(label2);
    }
}

```

```
JPanel panel2 = new JPanel(layout:null);
panel2.setBounds(x:0, y:0, width:90, height:600);
panel2.setBackground(Color.red);
frame.add(panel2);

JPanel panel = new JPanel(layout:null);
panel.setBounds(x:0,y:0,width:600,height:70);
panel.setBackground(Color.blue);
frame.add(panel);

JButton botonAdmin = new JButton(text:"Admin");
botonAdmin.setBounds(x:5, y:90, width:80, height:25);
panel2.add(botonAdmin);

JButton botonUser = new JButton(text:"User");
botonUser.setBounds(x:5, y:180, width:80, height:25);
panel2.add(botonUser);

JButton botonWorker = new JButton(text:"Worker");
botonWorker.setBounds(x:5, y:270, width:80, height:25);
panel2.add(botonWorker);
```

```
        botonAdmin.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                CLI cli = new CLI();
                cli.loginAdmin();
            }
        });

        botonUser.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                CLI_Client cli_Client = new CLI_Client();
                cli_Client.loginUser();
            }
        });

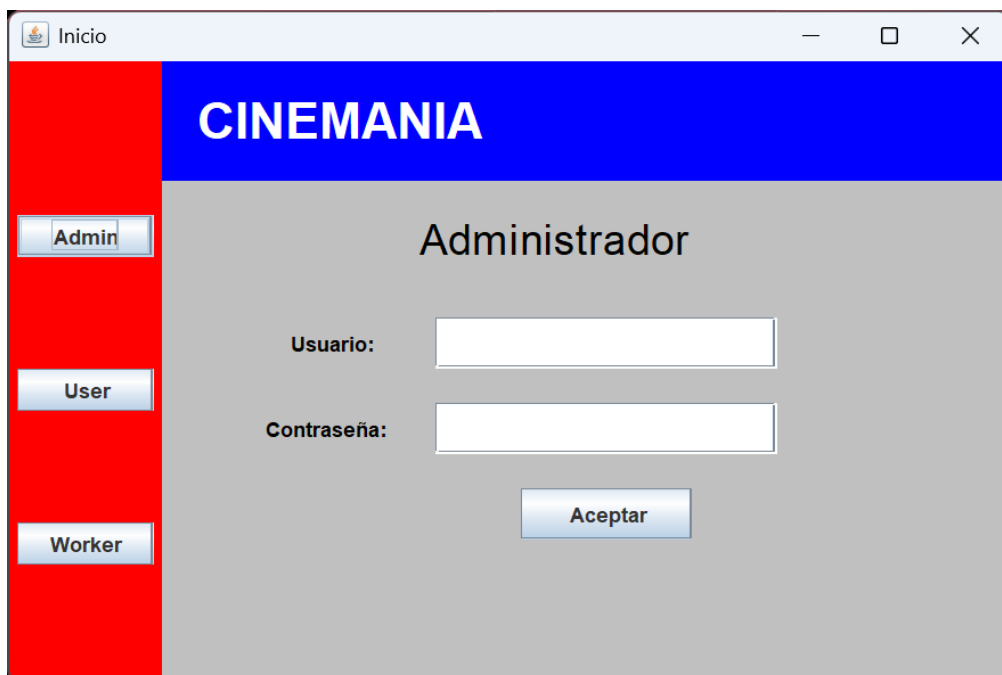
        botonWorker.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                CLI_Worker cli_Worker = new CLI_Worker();
                cli_Worker.loginWorker();
            }
        });

        frame.setVisible(b:true);
    }
}
```

```
package com.example;

import com.example.UI.*;
public class Main {
    /*corre el programa */
    Run | Debug
    public static void main(String[] args) {
        CLI2 cli2 = new CLI2();
        cli2.window();
    }
}
```

Resultado:



Inicio

CINEMANIA

Admin

User

Worker

Usuario

Usuario:

Contraseña:

Registrarse Aceptar

Inicio

CINEMANIA

Admin

User

Worker

Trabajador

Usuario:

ID:

Departamento:

Registrarse Aceptar

